

1. STRING LENGTH

```
#include
int main() {
char str[100];
int length = 0;
printf("Enter a string: ");
fgets(str, sizeof(str), stdin);
while (str[length] != '\0' && str[length] != '\n') {
length++;
}
printf("Length = %d\n", length);
return 0;
}
```

Input

Enter a string: Hello

Output

Length = 5

2. ALPHABETICAL ORDER OF NAMES

```
#include
#include
int main() {
int n, i, j;
char names[50][50], temp[50];
printf("Enter number of names: ");
scanf("%d", &n);
printf("Enter names:\n");
for (i = 0; i < n; i++) {
scanf("%s", names[i]);
}
for (i = 0; i < n - 1; i++) {
for (j = i + 1; j < n; j++) {
if (strcmp(names[i], names[j]) > 0) {
strcpy(temp, names[i]);
```

```
strcpy(names[i], names[j]);
strcpy(names[j], temp);
}
}
}
printf("Names in alphabetical order:\n");
for (i = 0; i < n; i++) {
printf("%s\n", names[i]);
}
return 0;
}
```

Input

Enter number of names: 4

Enter names:

Zara

Anu

Rahul

Bala

Output

Names in alphabetical order:

Anu

Bala

Rahul

Zara

3. PALINDROME

```
#include
```

```
#include
```

```
int main() {
```

```
char str[100];
```

```
int i, j, flag = 1;
```

```
printf("Enter a string: ");
```

```
scanf("%s", str);
```

```
i = 0;
```

```

j = strlen(str) - 1;
while (i < j) {
if (str[i] != str[j]) {
flag = 0;
break;
}
i++;
j--;
}
if (flag)
printf("Palindrome\n");
else
printf("Not Palindrome\n");
return 0;
}

```

Input

Enter a string: madam

Output

Palindrome

4. COUNT VOWELS, CONSONANTS AND SPECIAL CHARACTERS

```

#include
int main() {
char str[100];
int i, vowels = 0, consonants = 0, special = 0;
printf("Enter a string: ");
fgets(str, sizeof(str), stdin);
for (i = 0; str[i] != '\0'; i++) {
char ch = str[i];
if (ch == '\n')
continue;
if (ch == 'a' || ch == 'e' || ch == 'i' ||
ch == 'o' || ch == 'u' ||
ch == 'A' || ch == 'E' || ch == 'I' ||

```

```

ch == 'O' || ch == 'U') {
vowels++;
}
else if ((ch >= 'A' && ch <= 'Z') ||
(ch >= 'a' && ch <= 'z')) {
consonants++;
}
else {
special++;
}
}
printf("Vowels = %d\n", vowels);
printf("Consonants = %d\n", consonants);
printf("Special characters = %d\n", special);
return 0;
}

```

Input

Enter a string: Hello@123

Output

Vowels = 2

Consonants = 3

Special characters = 4

5. LOWERCASE TO UPPERCASE

```
#include
```

```

int main() {
char str[100];
int i;
printf("Enter a string: ");
fgets(str, sizeof(str), stdin);
for (i = 0; str[i] != '\0'; i++) {
if (str[i] >= 'a' && str[i] <= 'z') {
str[i] = str[i] - 32;
}
}
}

```

```
}  
printf("Uppercase string: %s", str);  
return 0;  
}
```

Input

Enter a string: hello world

Output

Uppercase string: HELLO WORLD

6. LARGEST OF THREE NUMBERS

```
#include  
int main() {  
    int a, b, c, largest;  
    printf("Enter three numbers: ");  
    scanf("%d %d %d", &a, &b, &c);  
    if (a >= b && a >= c)  
        largest = a;  
    else if (b >= a && b >= c)  
        largest = b;  
    else  
        largest = c;  
    printf("Largest = %d\n", largest);  
    return 0;  
}
```

Input

Enter three numbers: 25 70 45

Output

Largest = 70

7. ARMSTRONG NUMBER

```
#include  
int main() {  
    int number, temp, digit, sum = 0;  
    printf("Enter a number: ");  
    scanf("%d", &number);
```

```

temp = number;
while (temp > 0) {
digit = temp % 10;
sum = sum + digit * digit * digit;
temp = temp / 10;
}
if (sum == number)
printf("Armstrong Number\n");
else
printf("Not an Armstrong Number\n");
return 0;
}

```

Input

Enter a number: 153

Output

Armstrong Number

8. SUM OF DIGITS

```

#include
int main() {
int number, digit, sum = 0;
printf("Enter a number: ");
scanf("%d", &number);
while (number > 0) {
digit = number % 10;
sum = sum + digit;
number = number / 10;
}
printf("Sum of digits = %d\n", sum);
return 0;
}

```

Input

Enter a number: 12345

Output

Sum of digits = 15

9. ELECTRICITY BILL

Using:

First 100 units → ₹1.50/unit

101–200 units → ₹2.50/unit

Above 200 units → ₹4.00/unit

```
#include
```

```
int main() {
```

```
float units, bill;
```

```
printf("Enter units consumed: ");
```

```
scanf("%f", &units);
```

```
if (units <= 100) {
```

```
bill = units * 1.5;
```

```
}
```

```
else if (units <= 200) {
```

```
bill = 100 * 1.5 + (units - 100) * 2.5;
```

```
}
```

```
else {
```

```
bill = 100 * 1.5 + 100 * 2.5 + (units - 200) * 4.0;
```

```
}
```

```
printf("Electricity Bill = Rs. %.2f\n", bill);
```

```
return 0;
```

```
}
```

Input

Enter units consumed: 250

Output

Electricity Bill = Rs. 550.00

10. FIBONACCI SERIES

```
#include
```

```
int main() {
```

```
int n, i;
```

```
int a = 0, b = 1, c;
```

```
printf("Enter N: ");
```

```

scanf("%d", &n);
for (i = 1; i <= n; i++) {
printf("%d ", a);
c = a + b;
a = b;
b = c;
}
return 0;
}

```

Input

Enter N: 7

Output

0 1 1 2 3 5 8

11. SUM OF FIRST N NATURAL NUMBERS

```

#include
int main() {
int n, i, sum = 0;
printf("Enter N: ");
scanf("%d", &n);
for (i = 1; i <= n; i++) {
sum = sum + i;
}
printf("Sum = %d\n", sum);
return 0;
}

```

Input

Enter N: 10

Output

Sum = 55

12. SIMPLE INTEREST

```

#include
int main() {
float p, r, t, si;

```



```
printf("Enter Principal, Rate and Time: ");
scanf("%f %f %f", &p, &r, &t);
si = (p * r * t) / 100;
printf("Simple Interest = %.2f\n", si);
return 0;
}
```

Input

Enter Principal, Rate and Time: 10000 5 2

Output

Simple Interest = 1000.00

13. MULTIPLICATION TABLE

```
#include
int main() {
int n, i;
printf("Enter a number: ");
scanf("%d", &n);
for (i = 1; i <= 10; i++) {
printf("%d x %d = %d\n", n, i, n * i);
}
return 0;
}
```

Input

Enter a number: 5

Output

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

5 x 4 = 20

5 x 5 = 25

5 x 6 = 30

5 x 7 = 35

5 x 8 = 40

5 x 9 = 45

5 x 10 = 50

14. FIRST N EVEN NUMBERS

```
#include  
int main() {  
    int n, i;  
    printf("Enter N: ");  
    scanf("%d", &n);  
    for (i = 1; i <= n; i++) {  
        printf("%d ", 2 * i);  
    }  
    return 0;  
}
```

Input

Enter N: 5

Output

2 4 6 8 10

15. FACTORIAL

```
#include  
int main() {  
    int n, i;  
    long long fact = 1;  
    printf("Enter N: ");  
    scanf("%d", &n);  
    for (i = 1; i <= n; i++) {  
        fact = fact * i;  
    }  
    printf("Factorial = %lld\n", fact);  
    return 0;  
}
```

Input

Enter N: 5

Output

Factorial = 120

16. BIGGEST OF N NUMBERS

```
#include
int main() {
int n, i, num, max;
printf("Enter N: ");
scanf("%d", &n);
printf("Enter numbers:\n");
scanf("%d", &max);
for (i = 1; i < n; i++) {
scanf("%d", &num);
if (num > max)
max = num;
}
printf("Biggest = %d\n", max);
return 0;
}
```

Input

Enter N: 5

Enter numbers:

10 45 23 89 34

Output

Biggest = 89

17. POSITIVE AND NEGATIVE COUNT

```
#include
int main() {
int n, i, num;
int positive = 0, negative = 0;
printf("Enter N: ");
scanf("%d", &n);
printf("Enter numbers:\n");
for (i = 0; i < n; i++) {
scanf("%d", &num);
if (num > 0)
```

```

positive++;
else if (num < 0)
negative++;
}
printf("Positive numbers = %d\n", positive);
printf("Negative numbers = %d\n", negative);
return 0;
}

```

Input

Enter N: 6

Enter numbers:

10 -5 20 -8 -3 15

Output

Positive numbers = 3

Negative numbers = 3

18. SUM AND AVERAGE OF ARRAY

```

#include
int main() {
int n, i, a[100], sum = 0;
float average;
printf("Enter N: ");
scanf("%d", &n);
printf("Enter array elements:\n");
for (i = 0; i < n; i++) {
scanf("%d", &a[i]);
sum = sum + a[i];
}
average = (float)sum / n;
printf("Sum = %d\n", sum);
printf("Average = %.2f\n", average);
return 0;
}

```

Input

Enter N: 5

Enter array elements:

10 20 30 40 50

Output

Sum = 150

Average = 30.00

19. LINEAR SEARCH

```
#include
```

```
int main() {
```

```
int n, i, a[100], key, found = 0;
```

```
printf("Enter N: ");
```

```
scanf("%d", &n);
```

```
printf("Enter array elements:\n");
```

```
for (i = 0; i < n; i++) {
```

```
scanf("%d", &a[i]);
```

```
}
```

```
printf("Enter key: ");
```

```
scanf("%d", &key);
```

```
for (i = 0; i < n; i++) {
```

```
if (a[i] == key) {
```

```
found = 1;
```

```
break;
```

```
}
```

```
}
```

```
if (found)
```

```
printf("Found\n");
```

```
else
```

```
printf("Not Found\n");
```

```
return 0;
```

```
}
```

Input

Enter N: 5

Enter array elements:

10 20 30 40 50

Enter key: 30

Output

Found

20. SORT IN INCREASING ORDER

```
#include
```

```
int main() {
```

```
int n, i, j, a[100], temp;
```

```
printf("Enter N: ");
```

```
scanf("%d", &n);
```

```
printf("Enter array elements:\n");
```

```
for (i = 0; i < n; i++) {
```

```
scanf("%d", &a[i]);
```

```
}
```

```
for (i = 0; i < n - 1; i++) {
```

```
for (j = i + 1; j < n; j++) {
```

```
if (a[i] > a[j]) {
```

```
temp = a[i];
```

```
a[i] = a[j];
```

```
a[j] = temp;
```

```
}
```

```
}
```

```
}
```

```
printf("Sorted array:\n");
```

```
for (i = 0; i < n; i++) {
```

```
printf("%d ", a[i]);
```

```
}
```

```
return 0;
```

```
}
```

Input

Enter N: 5

Enter array elements:

50 20 40 10 30

Output

Sorted array:

10 20 30 40 50

21. SWAP TWO NUMBERS

```
#include
```

```
int main() {
```

```
int a, b, temp;
```

```
printf("Enter A and B: ");
```

```
scanf("%d %d", &a, &b);
```

```
temp = a;
```

```
a = b;
```

```
b = temp;
```

```
printf("After swapping:\n");
```

```
printf("A = %d\n", a);
```

```
printf("B = %d\n", b);
```

```
return 0;
```

```
}
```

Input

Enter A and B: 10 20

Output

After swapping:

A = 20

B = 10

22. ADDITION, SUBTRACTION, MULTIPLICATION, DIVISION

```
#include
```

```
int main() {
```

```
float a, b;
```

```
printf("Enter A and B: ");
```

```
scanf("%f %f", &a, &b);
```

```
printf("Addition = %.2f\n", a + b);
```

```
printf("Subtraction = %.2f\n", a - b);
```

```
printf("Multiplication = %.2f\n", a * b);
```

```
if (b != 0)
```

```
printf("Division = %.2f\n", a / b);  
else  
printf("Division is not possible\n");  
return 0;  
}
```

Input

Enter A and B: 20 5

Output

Addition = 25.00

Subtraction = 15.00

Multiplication = 100.00

Division = 4.00

23. CELSIUS TO FAHRENHEIT

```
#include  
int main() {  
float c, f;  
printf("Enter temperature in Celsius: ");  
scanf("%f", &c);  
f = (9 * c / 5) + 32;  
printf("Temperature in Fahrenheit = %.2f\n", f);  
return 0;  
}
```

Input

Enter temperature in Celsius: 25

Output

Temperature in Fahrenheit = 77.00

24. AREA AND PERIMETER OF RECTANGLE

```
#include  
int main() {  
float length, breadth, area, perimeter;  
printf("Enter length and breadth: ");  
scanf("%f %f", &length, &breadth);  
area = length * breadth;
```



```
perimeter = 2 * (length + breadth);  
printf("Area = %.2f\n", area);  
printf("Perimeter = %.2f\n", perimeter);  
return 0;  
}
```

Input

Enter length and breadth: 10 5

Output

Area = 50.00

Perimeter = 30.00

25. EVEN OR ODD

```
#include  
int main() {  
int number;  
printf("Enter a number: ");  
scanf("%d", &number);  
if (number % 2 == 0)  
printf("Even\n");  
else  
printf("Odd\n");  
return 0;  
}
```

Input

Enter a number: 24

Output

Even

26. DUPLICATE ELEMENTS IN ARRAY

```
#include  
int main() {  
int n, a[100], i, j;  
printf("Enter N: ");  
scanf("%d", &n);  
printf("Enter array elements:\n");
```

```

for (i = 0; i < n; i++) {
scanf("%d", &a[i]);
}
printf("Duplicate elements:\n");
for (i = 0; i < n; i++) {
for (j = i + 1; j < n; j++) {
if (a[i] == a[j]) {
printf("%d ", a[i]);
break;
}
}
}
return 0;
}

```

Input

Enter N: 6

Enter array elements:

10 20 30 20 40 10

Output

Duplicate elements:

10 20

27. MATRIX ADDITION

```
#include
```

```

int main() {
int a[10][10], b[10][10], sum[10][10];
int rows, cols, i, j;
printf("Enter rows and columns: ");
scanf("%d %d", &rows, &cols);
printf("Enter first matrix:\n");
for (i = 0; i < rows; i++) {
for (j = 0; j < cols; j++) {
scanf("%d", &a[i][j]);
}
}

```

```

}
printf("Enter second matrix:\n");
for (i = 0; i < rows; i++) {
    for (j = 0; j < cols; j++) {
        scanf("%d", &b[i][j]);
    }
}
for (i = 0; i < rows; i++) {
    for (j = 0; j < cols; j++) {
        sum[i][j] = a[i][j] + b[i][j];
    }
}
printf("Result matrix:\n");
for (i = 0; i < rows; i++) {
    for (j = 0; j < cols; j++) {
        printf("%d ", sum[i][j]);
    }
    printf("\n");
}
return 0;
}

```

Input

Enter rows and columns: 2 2

Enter first matrix:

1 2

3 4

Enter second matrix:

5 6

7 8

Output

Result matrix:

6 8

10 12

28. MATRIX MULTIPLICATION

```
#include
int main() {
int a[10][10], b[10][10], result[10][10];
int r1, c1, r2, c2;
int i, j, k;
printf("Enter rows and columns of first matrix: ");
scanf("%d %d", &r1, &c1);
printf("Enter rows and columns of second matrix: ");
scanf("%d %d", &r2, &c2);
if (c1 != r2) {
printf("Matrix multiplication not possible\n");
return 0;
}
printf("Enter first matrix:\n");
for (i = 0; i < r1; i++) {
for (j = 0; j < c1; j++) {
scanf("%d", &a[i][j]);
}
}
printf("Enter second matrix:\n");
for (i = 0; i < r2; i++) {
for (j = 0; j < c2; j++) {
scanf("%d", &b[i][j]);
}
}
for (i = 0; i < r1; i++) {
for (j = 0; j < c2; j++) {
result[i][j] = 0;
for (k = 0; k < c1; k++) {
result[i][j] += a[i][k] * b[k][j];
}
}
}
```

```

}
printf("Result matrix:\n");
for (i = 0; i < r1; i++) {
for (j = 0; j < c2; j++) {
printf("%d ", result[i][j]);
}
printf("\n");
}
return 0;
}

```

Input

Enter rows and columns of first matrix: 2 2

Enter rows and columns of second matrix: 2 2

Enter first matrix:

1 2

3 4

Enter second matrix:

5 6

7 8

Output

Result matrix:

19 22

43 50

29. MATRIX TRANSPOSE

```
#include
```

```

int main() {
int matrix[10][10], transpose[10][10];
int rows, cols, i, j;
printf("Enter rows and columns: ");
scanf("%d %d", &rows, &cols);
printf("Enter matrix:\n");
for (i = 0; i < rows; i++) {
for (j = 0; j < cols; j++) {

```

```

scanf("%d", &matrix[i][j]);
}
}
for (i = 0; i < rows; i++) {
for (j = 0; j < cols; j++) {
transpose[j][i] = matrix[i][j];
}
}
printf("Transpose:\n");
for (i = 0; i < cols; i++) {
for (j = 0; j < rows; j++) {
printf("%d ", transpose[i][j]);
}
printf("\n");
}
return 0;
}

```

Input

Enter rows and columns: 2 3

Enter matrix:

1 2 3

4 5 6

Output

Transpose:

1 4

2 5

3 6

30. SUM OF DIAGONAL ELEMENTS

```
#include
```

```
int main() {
```

```
int matrix[10][10];
```

```
int n, i, j, sum = 0;
```

```
printf("Enter size of square matrix: ");
```

```
scanf("%d", &n);
printf("Enter matrix:\n");
for (i = 0; i < n; i++) {
    for (j = 0; j < n; j++) {
        scanf("%d", &matrix[i][j]);
    }
}
for (i = 0; i < n; i++) {
    sum = sum + matrix[i][i];
}
printf("Sum of diagonal elements = %d\n", sum);
return 0;
}
```

Input

Enter size of square matrix: 3

Enter matrix:

1 2 3

4 5 6

7 8 9

Output

Sum of diagonal elements = 15